

FAI Network Topology Governance

Derivation Note D2.61 — Series D, Phase D2, Note #556

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This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

When many CKS-governed Selves each configure their participation in Full Aspect Integration (FAI) events, the aggregate of those configurations produces a FAI network topology: a structured pattern of who connects to whom, at what depth, and in which direction. This note formalizes the prior-art claim that FAI network topology is a distributed governance phenomenon — it is not planned by any central authority but emerges from each Self's home governance substrate content. The note defines the four dimensions of topology (connectivity, directionality, density, and governance depth), identifies four architecturally possible topology patterns (mesh, hub-and-spoke, cluster, and chain) in the calibrated-humility register appropriate to population-scale architectural claims, states the governance conditions that could enable each pattern, summarizes the implications of each pattern for governance capacity and pattern propagation, and identifies topology lock-in as an anti-pattern that violates exit rights preservation.

1. Positioning and derivation

D2.61 operates at the intersection of two parent commitments. D1.26 establishes joint authority across population-level governance perimeters — the claim that as FAI events accumulate across many Selves, the collective architecture remains under distributed human governance rather than under any emerging central authority. D2.31 establishes the dimensions through which each individual Self governs its own FAI participation: its propagation permissions, trust calibrations, partner preferences, and network role configuration. D2.61 extends that individual-Self picture to the population level by asking what structure the aggregate of many Selves' participation configurations produces, and whether that structure is itself governable.

The answer this note formalizes is yes on both counts. The aggregate structure — the topology — exists as a real architectural object: it determines which Selves can exchange substrate content with which others, how quickly governance patterns can propagate across the network, and what governance attention burdens each Self carries. And the topology is governable in a specific sense: each Self can observe its own position within it and can adjust that position by modifying its participation configuration, which is home governance substrate content. No central authority needs to — or could — govern the topology from above. The governance is distributed by design.

This is a Claim 6 operational decomposition in the calibrated-humility register. Population-scale FAI networks do not yet exist as deployed systems. The architectural commitments described here state what the inheritance of the CKS trilogy's five prior claims produces at population scope; specific topological dynamics await empirical observation.

2. What FAI network topology means

In the FAI context, topology names the structural-arrangement property of a population-scale network of CKS-governed Selves — which Selves have established FAI relationships with which others, and the character of each relationship. Topology is not borrowed from graph theory or network-science terminology in any technical sense; it names the architectural arrangement that governs how substrate content, governance patterns, and coordination provenance can flow across organizational and domain boundaries.

A FAI network topology has four dimensions:

Connectivity names which Selves have established active FAI relationships with which others. Two Selves are connected if their participation configurations have established a FAI relationship — meaning each Self's governance has authorized the other as a recognized FAI partner and has configured sharing scope for the relationship. Selves that have not established such mutual configuration are unconnected in the topology, regardless of their proximity in any other sense (same domain, same organization, same infrastructure).

Directionality names the role each Self plays within its established relationships. A Self configured as an initiating-role participant introduces FAI events — contributes aspects to the shared substrate and drives event formation. A Self configured as a responding-role participant receives and processes FAI events initiated by others. A Self configured in both roles participates symmetrically. Directionality is not fixed by the architecture; it is governance-configured within each Self's participation configuration and can differ relationship by relationship.

Density names how many active FAI relationships each Self maintains. A Self with many active FAI relationships occupies a high-density position; a Self with few relationships occupies a low-density position. Density is a consequence of each Self's partner preference configuration and trust calibrations, not an architectural constraint. The architecture does not

impose a minimum or maximum on relationship count.

Governance depth names the richness of each established relationship. A minimal relationship may be single-aspect with low-trust configuration and shallow provenance carry-over. A deep relationship may be multi-aspect, high-trust, with rich standing configurations and deep provenance carry-over depth. Governance depth for a given relationship is determined jointly by the participating Selves' governance authorities, since FAI is a joint governance commitment by architectural property. Depth can vary substantially across different relationships maintained by the same Self.

These four dimensions together characterize any specific topology a population-scale FAI network could exhibit.

3. Topology as distributed governance phenomenon

The central prior-art claim this note formalizes is that FAI network topology is not planned by any central authority. No coordinating entity, platform operator, or governance body determines which Selves connect to which others, at what density, or with what governance depth. The topology emerges entirely from each Self's governance decisions about its own participation configuration.

Each Self's participation configuration is home governance substrate content — authored under the authority of that Self's human governance, subject to all six Paper 1 commitments (inspect, modify, override, at any time, without approval). When a Self's governance authorities decide to add a FAI relationship with a new partner, modify the trust calibration for an existing relationship, restrict propagation permissions in response to a governance concern, or expand its network role from responding-only to initiating-and-responding, these decisions are substrate edits under home authority. They take effect as changes to the Self's participation configuration.

The topology of the network as a whole is the aggregate of all such substrate edits across all participating Selves. It is a governance-distributed emergence in a precise sense: no single governance authority holds the authority to determine the global topology; the global topology is what you get when you read out the connection pattern implied by all Selves' participation configurations simultaneously.

This has an important implication for governance practitioners in a population-scale network. A Self's governance authorities can observe their Self's current network position — how many active FAI relationships it has, with whom, at what depth — by reading their own participation configuration and governance trust records. They cannot directly observe the full global topology, because other Selves' participation configurations are home governance content under those Selves' authority. But they can observe and govern their own position within it. This is the governance-distributed topology governance the note formalizes.

4. Four architecturally possible topology patterns

The architecture does not prescribe how the topology should be organized. The calibrated-humility register appropriate to Claim 6 territory requires treating what follows as architecturally possible patterns — patterns the architecture enables and that could emerge from particular distributions of governance decisions — not as prescriptions, predictions, or idealized forms. Different deployment contexts may produce different topologies; the same deployment may transition between patterns over time as participation configurations change.

Pattern 1 — Mesh topology

In a mesh topology, many Selves each maintain FAI relationships with many others. Connectivity is broadly distributed; no Self occupies a structurally privileged position by virtue of having uniquely many connections. Directionality tends toward symmetry in a mature mesh — many Selves configured in both initiating and responding roles. Density is high across much of the network.

A mesh topology is architecturally possible where governance trust is broadly distributed and propagation permissions are configured generously. Where many Selves have established mutual trust calibrations with many partners and have configured sharing scope that permits broad content flow, the connection pattern that emerges can exhibit mesh-like density.

Governance implications: a mesh topology maximizes pattern propagation potential — substrate-content patterns such as schemas, orchestration substrates, and protocol logics can travel across many paths between any two Selves, rather than depending on any single relay. Against this, the governance attention requirement per Self is high. Each Self must maintain trust calibrations, partner preferences, and conflict-handling configurations across many active relationships. The governance cost of maintaining a high-density position is borne by each Self's governance authorities; the architecture does not reduce this cost by centralizing it.

Pattern 2 — Hub-and-spoke topology

In a hub-and-spoke topology, a minority of Selves act as FAI hubs — maintaining many connections, configured in both initiating and responding roles, with rich governance depth across their relationships. The majority of Selves maintain few connections, often configured primarily in the responding role, relying on hub Selves to relay governance patterns across the broader network.

A hub-and-spoke topology is architecturally possible where some Selves configure broad network roles and generous partner preferences while others configure narrow roles — perhaps because their governance authorities have chosen to limit the governance attention burden that high-density participation would impose. Hub positions emerge from governance decisions, not from architectural privilege; a Self becomes a hub because its participation configuration opens many relationships, not because any central authority designates it as one.

Governance implications: a hub-and-spoke topology concentrates pattern propagation capacity at hub Selves. Substrate-content patterns that reach a hub can propagate quickly to all of that hub's spokes. This concentration is also a governance attention concentration — hub Selves' governance authorities carry disproportionate governance burdens relative to spoke Selves. Whether that concentration is appropriate depends on the governance resources available at hub positions and on whether hub positions are genuinely voluntary — which returns to the exit rights commitment addressed in §5.

Pattern 3 — Cluster topology

In a cluster topology, Selves group into identifiable clusters — dense connection patterns within clusters, sparse connection patterns between them. Clusters may form along domain-specific lines, organizational boundaries, or governance-trust affinity lines. Within-cluster governance depth tends to be high; between-cluster relationships, where they exist, may be shallower or more restricted in sharing scope.

A cluster topology is architecturally possible where domain-selective participation is the dominant governance pattern. Where each Self's partner preferences strongly favor Selves in the same domain or organizational context, and where propagation permissions are more restrictive toward out-of-cluster Selves, the connection pattern that emerges exhibits cluster structure. The cluster boundaries are not architecturally imposed; they are the aggregate consequence of many Selves each governing their own participation conservatively toward unfamiliar partners.

Governance implications: a cluster topology preserves domain-specific specialization well. Substrate-content patterns that are strategically valuable within a cluster can develop without propagating outside it. Cross-cluster coordination, where it occurs at all, can be mediated through the sparse between-cluster relationships, allowing deliberate governance over what flows across cluster boundaries. The risk is isolation — clusters that develop no between-cluster relationships cannot benefit from coordination-substrate patterns that have matured in other parts of the network.

Pattern 4 — Chain topology

In a chain topology, Selves connect in sequential patterns — Self A connected to Self B, B connected to Self C, and so on — with few or no shortcuts across the chain. Governance patterns propagate along chains hop by hop. The topology has low density except at the local level and exhibits asymmetric directionality — each Self in the chain may play an initiating role toward the next Self while playing a responding role toward the prior one.

A chain topology is architecturally possible where trust calibrations are highly asymmetric in their formation patterns — where each Self's trust calibrations produce exactly one or two authoritative partners, and where governance maturity develops sequentially rather than broadly. A chain can also emerge where organizational structures produce sequential authority

relationships that governance practitioners replicate in their participation configurations.

Governance implications: a chain topology produces slow propagation velocity for substrate-content patterns — patterns must traverse each hop in sequence, with each hop governed by that relay Self's ingestion policies. Chains are fragile: the dissolution of any single FAI relationship in the chain interrupts the propagation path between the Selves on either side of the gap. Governance practitioners in a chain-structured network have strong incentives to develop redundant connection paths where governance resources permit, moving the topology toward mesh or cluster patterns at the local level.

5. Anti-pattern: topology lock-in

The four patterns above describe topologies that could emerge from governance decisions and could change as governance decisions change. The topology lock-in anti-pattern names the failure mode in which a Self cannot change its network topology — cannot add new FAI relationships, cannot dissolve existing ones, cannot adjust its governance depth or network role — due to governance arrangements that have eliminated its ability to do so.

Topology lock-in is an exit rights violation (D2.46) surfaced at the topology level. The exit rights commitment holds that each Self must retain the ability to withdraw from any FAI relationship, adjust its participation configuration, and change its network position. A governance arrangement that makes any of these unavailable — by committing a Self to permanent FAI relationships, by embedding topology configuration in non-substrate infrastructure that is not subject to the three rights (inspect, modify, override), or by creating contractual or operational dependencies that make topology change prohibitively costly — violates the exit rights commitment regardless of how the arrangement was formed.

The topology lock-in anti-pattern can appear in several operational forms. A Self whose FAI relationships are configured at the infrastructure level rather than as home governance substrate content cannot exercise its modify and override rights over its own topology. A Self whose participation configuration has been delegated to an external platform in ways that remove home governance authority faces the same limitation. A Self that has accepted governance arrangements conditioning the dissolution of one FAI relationship on the consent of a third party has effectively transferred exit rights to that third party.

The architectural remedy for each form is the same: FAI relationships and participation configurations must be maintained as home governance substrate content, subject to all six Paper 1 commitments, with no intermediary layer that can in principle prevent the Self's governance authorities from inspecting, modifying, or overriding them.

6. Operational test

In the calibrated-humility register appropriate to this note's Claim 6 territory, the operational test asks whether the architectural commitments are visible at the level of home governance

substrate content — whether the topology-governance properties can be determined by inspection rather than by inference or trust.

For a given Self in a population-scale FAI network, a governance observer with appropriate access should be able to determine:

1. Current network position. How many active FAI relationships does this Self maintain? By reading the Self's participation configuration (home governance substrate content), the observer can identify each established partner, the sharing scope for each relationship, the trust calibration for each partner, and the Self's configured network role. The count of active relationships and the identity of each partner are directly readable.

2. Relationship depth. What is the governance depth of each active relationship? By reading the per-relationship configuration — aspects authorized for sharing, provenance carry-over depth, conflict-handling configuration — the observer can characterize each relationship as minimal, moderate, or deep. This characterization is readable from substrate content without requiring access to the partner Self's home substrate.

3. Exit rights integrity. Can this Self modify its network topology by governance choice alone? By examining whether FAI relationship configurations reside as home governance substrate content subject to all three rights, and whether any configuration has been externalized to non-substrate infrastructure or made subject to third-party consent, the observer can determine whether topology lock-in conditions are present.

A Self that passes all three checks instantiates topology governance in the sense this note formalizes. A Self that fails check 3 is in topology lock-in regardless of whether it currently wishes to change its position.

7. Conclusion

FAI network topology is a distributed governance phenomenon: the structural pattern of connections across a population-scale CKS-governed Self network emerges from each Self's governance decisions about its own participation configuration, not from any central planning authority. This is the prior-art claim this note formalizes. Four topology patterns — mesh, hub-and-spoke, cluster, and chain — are architecturally possible within this distributed-emergence structure, each enabled by different distributions of governance decisions across the population. None is prescribed; each carries distinct implications for governance attention burden and pattern propagation velocity. The topology lock-in anti-pattern identifies the failure mode in which exit rights are violated at the topology level, and the operational test confirms that topology governance properties are readable from home governance substrate content by observers with appropriate access.

Subsequent derivation notes in the D2 phase will develop further operational decompositions of Claim 6 commitments. The calibrated-humility voice is maintained throughout: the architectural

inheritance of the CKS trilogy's five prior claims determines what is possible at population scope; the specific dynamics of any actual population-scale deployment await empirical observation.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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